PennyPal — Pro Code Navigator & Live Defense Guide

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Purpose: Jump directly to any screen's code in 1 second during live viva defense and explain it like a senior developer.

Rule #1: Never Browse Folders with Your Mouse

Real developers do not click through folders on the left sidebar. They use keyboard shortcuts.

When a judge asks: "*Can you show the code for this screen or feature?*"

1. Press **Ctrl + P** in VS Code (Quick Open).
2. Type 3 to 5 letters of the file name (for example: **home_sc** or **budget_sc**).
3. Press **Enter**. The file opens instantly in 1 second.
4. If asked for a specific word or function, press **Ctrl + F** and type the keyword.

Doing this gives judges immediate confidence that you wrote the codebase and know every file by name.

Quick Jump Table: What to Type in Ctrl + P

Screen / Feature	Type in Ctrl + P	File Path	What to Point to on Screen
Main Dashboard & Header	`home_screen`	`lib/features/dashboard/presentation/home_screen.dart`	`_HomeHeader` (Avatar & Bell), `_BalanceHero`, `_PaceRing`
Monthly Budget & 1-Tap Copy	`budget_screen`	`lib/features/budgets/presentation/budget_screen.dart`	`_copyPreviousMonthBudget()` method
Add Expense & Gemini AI OCR	`transaction_form`	`lib/features/transactions/presentation/transaction_form_screen.dart`	`_scanReceiptWithGemini()` or `_pickReceiptImage()`
Offline Cache Engine	`main.dart`	`lib/main.dart` (lines 62–65)	`persistenceEnabled: true` & `CACHE_SIZE_UNLIMITED`
Savings Goals & Milestones	`savings_screen`	`lib/features/savings/presentation/savings_screen.dart`	`_GoalCard` & progress calculation
Split Bill	`cards_screen`	`lib/features/cards/presentation/cards_screen.dart`	Bill splitting calculation and WhatsApp share
Borrow & Lend (Debts)	`debt_tracker`	`lib/features/debts/presentation/debt_tracker_screen.dart`	`_DebtCard` (Lent vs Borrowed)
Subscriptions	`subscriptions_screen`	`lib/features/subscriptions/presentation/subscriptions_screen.dart`	Monthly recurring list & total cost calculation
Learning Hub & Lessons	`learning_screen`	`lib/features/learning/presentation/learning_screen.dart`	Lesson cards and categories
AI Assistant (Gemini Chat)	`assistant_screen`	`lib/features/assistant/presentation/assistant_screen.dart`	Chat message stream & Gemini prompt handler
Admin Panel	`admin_dashboard`	`lib/features/admin/presentation/admin_dashboard_screen.dart`	Admin statistics cards & charts
Navigation & Route Guards	`router.dart`	`lib/app/router.dart`	Role check: student vs admin redirection

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How Our Code is Structured (The 10-Second Architecture Pitch)

If judges ask: "*How did you structure your codebase?*", say:

> "*We structured PennyPal using **Feature-Driven Architecture**.*
> *Everything is inside `lib/features/`. Each feature has its own folder with:*
> • `presentation/` *for UI screens and widgets.*

> • `domain/` *for models and financial calculations.*
> • `data/` *for Firebase repositories and API calls."*

What to Say While Opening Code (Short & Natural)

Speak these simple sentences while your fingers type the shortcut:

- *"Sure, let me open that file directly."* (press **Ctrl + P**)
- *"Here is our presentation widget."*
- *"We separated UI from business logic."*
- *"State is handled through Riverpod providers."*
- *"Here is the calculation function."*

The 3 Most Likely Code Questions & How to Answer

Q1: "Show me how Offline Mode works in the code."

1. Press **Ctrl + P** ➔ type `main.dart` ➔ press **Enter**.
2. Scroll to lines 62–65:

```
FirebaseFirestore.instance.settings = const Settings(
  persistenceEnabled: true,
  cacheSizeBytes: Settings.CACHE_SIZE_UNLIMITED,
);
```

3. **What to say:**

> *"Right here in `main.dart`, we configure Firestore with `persistenceEnabled: true` and unlimited local cache. So every transaction writes to local device storage first, and syncs automatically when internet returns."*

Q2: "Show me where you call Gemini AI for receipts."

1. Press **Ctrl + P** ➔ type `transaction_form` ➔ press **Enter**.
2. Press **Ctrl + F** ➔ type `gemini` or `scan`.
3. **What to say:**

> *"Here in `transaction_form_screen.dart`, when the user picks a receipt photo, we send the image bytes to Gemini Vision AI. It extracts the merchant name, date, and amount, and populates the text controllers automatically."*

Q3: "Show me where the 1-Tap Quick Copy logic is."

1. Press **Ctrl + P** ➔ type `budget_screen` ➔ press **Enter**.
2. Press **Ctrl + F** ➔ type `copy`.
3. **What to say:**

> *"Here in `budget_screen.dart`, this function reads last month's budget documents from Firestore and creates duplicates for the current calendar month with one tap."*

Quick Tips for Live Confidence

- **Practice 3 times now:** Press **Ctrl + P** and jump between `home_screen.dart`, `budget_screen.dart`, and `main.dart`.
- **Stay relaxed:** If you misspell a file name, just press **Backspace** and retype. Never panic.
- **Point with your finger or mouse:** Once the file opens, highlight the class name or function with your mouse so the judge's eyes follow your lead.

🗣️ Natural Fillers & Transition Phrases (Spoken Cheat Sheet)

Use these short, easy-to-pronounce phrases whenever you switch screens, tap buttons, or open code. They fill pauses naturally without saying "umm" or "uhh":

1. Moving to the Next Screen or Feature

- *"Now next, we have..."*
- *"Let's jump into the [Screen Name]..."*
- *"Moving forward to our next screen..."*
- *"From here, let's head over to..."*
- *"Next up, let me show you..."*
- *"Now, switching over to the..."*
- *"Let's take a look at our next key feature..."*

2. Introducing a Screen or UI Component

- "*This is our [XYZ] screen.*"
- "*Right here on this screen, you can see...*"
- "*As you can see right in front of you...*"
- "*At first glance on this dashboard...*"
- "*This section is specifically built for...*"
- "*Here is where the student spends most of their time...*"
- "*Right at the top header, notice...*"

3. Demonstrating a Live Action

- "*Now, watch what happens when I tap this...*"
- "*Let me perform a live test right now...*"
- "*Notice how quickly this updates on screen...*"
- "*Let's see this in action...*"
- "*Just like that, it saves with zero delay.*"
- "*Watch this: I will add a test transaction...*"
- "*Now observe how the numbers adjust immediately...*"

4. Opening & Pointing to Code

- "*Sure, let's jump right into the code for this.*"
- "*Let me pull up the exact file for you.*"
- "*Here is the exact method handling this logic.*"
- "*Notice this function right here on line [X]...*"
- "*This is where the calculation happens under the hood.*"
- "*As you can see in this controller...*"
- "*We kept this logic completely separate from the UI.*"

5. Buying 2 to 3 Seconds to Think (Zero Panic)

- "*That is a very good question.*"
- "*Let me explain the exact reason behind this...*"
- "*To put this in simple words...*"
- "*The main idea here was...*"
- "*Let me show you this directly on screen...*"
- "*From a practical student point of view...*"

6. Wrapping Up a Feature & Moving Ahead

- "*And that is basically how this works.*"
- "*So, that wraps up this part.*"
- "*Now, with that in place, let's look at...*"
- "*That completes the student side of this feature.*"